When Generated Workflows Help Temporal Graph Repair: Verification Budgets and Merge Safety

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Abstract

An operational history can lose evidence through missing links, wrong joins or fragmented identities. Generated workflows provide exact repair references, but their value must be tested against repeated corruption of the same verified real histories and against the errors introduced by repair. We compare eight real-history corruption views with the same data plus executable generated workflows at matched simulated verification budgets, including fitting and calibration. An exploratory study supplies input ablations and fixed models; a prospective confirmation evaluates 36 unchanged fits in 768 conditions on 1,024 untouched root histories per source, using public loan and purchasing logs with controlled injected faults. At the prespecified 64-bundle loan setting with a .99 calibration precision target, generated training raises mean fully recovered events from 30.2 to 62.5 per 128-root block and reduces newly damaged events from 2.08 to 0.00. The paired recovery gain is 32.3 [29.5, 35.3] under conditional 95% block-bootstrap intervals, although one of three fitted-seed means loses recovery and mean history F1 falls by 0.23 percentage points. At 256 bundles, augmentation has worse mean history F1, recovery and damage in both sources at both tested targets. Merge guards substantially reduce harm relative to unguarded decoding, but a never-merge baseline exposes their residual risk: no tested source/training combination gains from merging without added harm. These results establish conditional value for generated supervision and quantify the cost of identity recovery. They do not establish natural-error performance, human verification savings or automatic-repair safety.

1 Application and contribution

A cancellation can be correctly recorded yet absent from the history retrieved for an order. A wrong join can make another customer's record appear in that history. When two systems assign different identifiers to the same object, either identifier may retrieve only part of its evidence. These failures make operational investigation difficult even when activities and timestamps themselves are accurate.

The practical objective is to restore recorded evidence, including exceptional behavior. We retain cancellations, returns and repeated work rather than changing the record to resemble an ideal process. The input is an available window of event–object relations, and the output is an auditable set of association and identity edits. This is offline reconstruction; later observations inside the window can support an earlier association.

Training such a repair system requires examples of correct histories and the ways their observed connections can fail. Verification is costly because an investigator may need to reconcile several records and identifiers. Executable workflows offer exact symbolic references, but simulated observations can contain misleading regularities. For example, an amount may be constant throughout a simulated order while the corresponding operational measurement changes between activities. A learner can mistake such differences for incorrect joins.

We ask two linked application questions. Do generated workflow histories add value beyond repeated corruption of the same verified operational histories? Do conservative identity merges recover useful evidence beyond a policy that never merges? Both questions count labels used for calibration and measure newly introduced errors. Improved average retrieval is insufficient if a method introduces damaging false associations.

The contribution is a controlled evaluation protocol for coupled temporal association and identity repair. It compares generated supervision with repeated corruption of the same verified histories, removes temporal, relational and descriptive evidence before retrieval and refitting, and tests identity safeguards against complete merge abstention. Untouched histories then test the exploratory findings with fixed models and thresholds. The classifiers, observation-randomization principle and graph operators are established components; the contribution is the evidence about their value and failure conditions under explicit verification and harm accounting.

2 Related work and comparison scope

HERE combines a Transformer with optional expert knowledge for missing case identifiers [1]. We retain its author model, tensor dataset and greedy decoder in a separate single-case comparison. Its output contract does not directly express simultaneous multi-object participation edits and alias merging. SANAGRAPH repairs event-log attributes using supplied trace structure [2]; providing correct trace membership would expose a target of our task. DDTR and process-tree approaches recover trace content [3, 4]. Their objectives differ from preserving event content while restoring its associations.

Contextual Event Knowledge Graph inference uses available relations and physical rules to infer missing entity identifiers [8]. Object-centric imperfection patterns identify association and identity problems encountered in practice [9]. Their public artifacts include injected event–object faults and observed cross-log object–object disagreements. The latter are useful evidence of data inconsistency, but do not independently certify the missing or wrong event–object links evaluated here. We do not claim to introduce these error categories.

Object-centric simulation [7], configurable process-data faults [10], and temporal graph generation [14] supply relevant building blocks. Domain randomization is an established transfer strategy [11]; synthetic relational pretraining also has recent precedents [12, 13]. Our question concerns what these ideas buy an operational repair application under explicit information and verification limits. HoloClean illustrates joint evidence-based data repair [5]; the present study concentrates on temporal histories, their identity fragmentation and the consequences of repairing them.

3 Task and executable supervision

3.1 Observed information and permitted edits

An event is $e = (u, t, a, P, x)$: an opaque handle, time, activity, typed participants and legitimate descriptive fields. Event–object incidence and temporal adjacency within observed histories form the temporal graph. Equal-time records remain simultaneous; source row order supplies no causal ordering. A typed catalog contains the available object handles, including any aliases.

The system can add or remove participation and merge same-type aliases. It cannot invent events, change recorded activities or times, create entirely unknown entities, or split an already overmerged identity. A merge consolidates handles and deduplicates participation. These capabilities matter when choosing a dataset: natural errors requiring a different operation cannot validate

this repair vocabulary.

Only the observed graph and target-type schema enter inference. The canonical identity map, clean case containers, corruption locations, seeds and clean-derived adjacency remain outside the learner interface. Target links are corrupted while resources and, where available, vendors remain context. This is a scoped fault model, not evidence of robustness to unreliable timestamps or pervasive context corruption. Public logs provide reference representations for controlled fault injection; they are not certifications that every underlying business record is correct.

3.2 Generate execution, observation and recording faults separately

Executable worlds contain parents, work items, resources and optional batches. Mechanisms include branching, joins, resource queues, handoffs, multistage routes, repeated work, cancellations and returns. Queueing configurations enforce resource capacity. The worlds are symbolic workflows, not calibrated digital twins of the evaluated organizations.

Observation projection determines visible entity types and fields. Recording faults are applied afterwards, producing learner observations and separate participation/identity references. Normalized recipes, random states, source fingerprints and export hashes record provenance. Execution checkpoints reproduce uninterrupted runs. Observations sharing an execution stay in the same data split.

The earlier paired observation comparison holds executed events, timestamps, target types and faults fixed. Generated views retain original fields or introduce activity-dependent categories, noisy numerical measurements, incidental categories and time bands, with varying field availability. Measurements can depend on an observed amount and activity, but never on a target identifier. Clean and damaged versions receive aligned descriptions through builder-only execution ancestry. Each trial contains 512 fitting worlds and 96 generated calibration worlds. Three independently generated collections test sensitivity to generated training data.

The mixed fault family splits eligible histories at a temporal cut with probability .25, removes approximately .10 of each resulting handle's links in a contiguous block, and replaces surviving target links with probability .10 by a different same-type identity. Histories require at least four events to be fragmented. Missing-block widths use stochastic rounding so short histories can also lose links. A separately implemented pipeline stressor uses activity-routed aliases, a recording outage and persistent stale joins. These are declared experimental conditions, not estimates of natural incident rates.

3.3 Learn and apply auditable edits

Candidate retrieval uses temporal proximity, activity, co-participation and descriptive agreement. Participation features summarize history degree, temporal bracketing, transitions and contextual agreement. The query event is excluded from the evidence supporting its existing association. Identity features compare temporal boundaries and activity, context and field profiles. Opaque identifier values are not numerical predictors.

Separate boosted classifiers score additions, removals and merges. An addition is labelled correct when it supplies a missing reference entity; a removal when the current entity is incorrect; a merge when both handles refer to the same entity. Earlier generated training includes partial teacher-repaired states; the repeated-corruption models use initial corrupted states. Local targets do not guarantee improvement in final history retrieval. Earlier neural controls use the same candidates with MLP, GraphSAGE [6] or temporal message-passing scorers; architecture superiority is not our contribution.

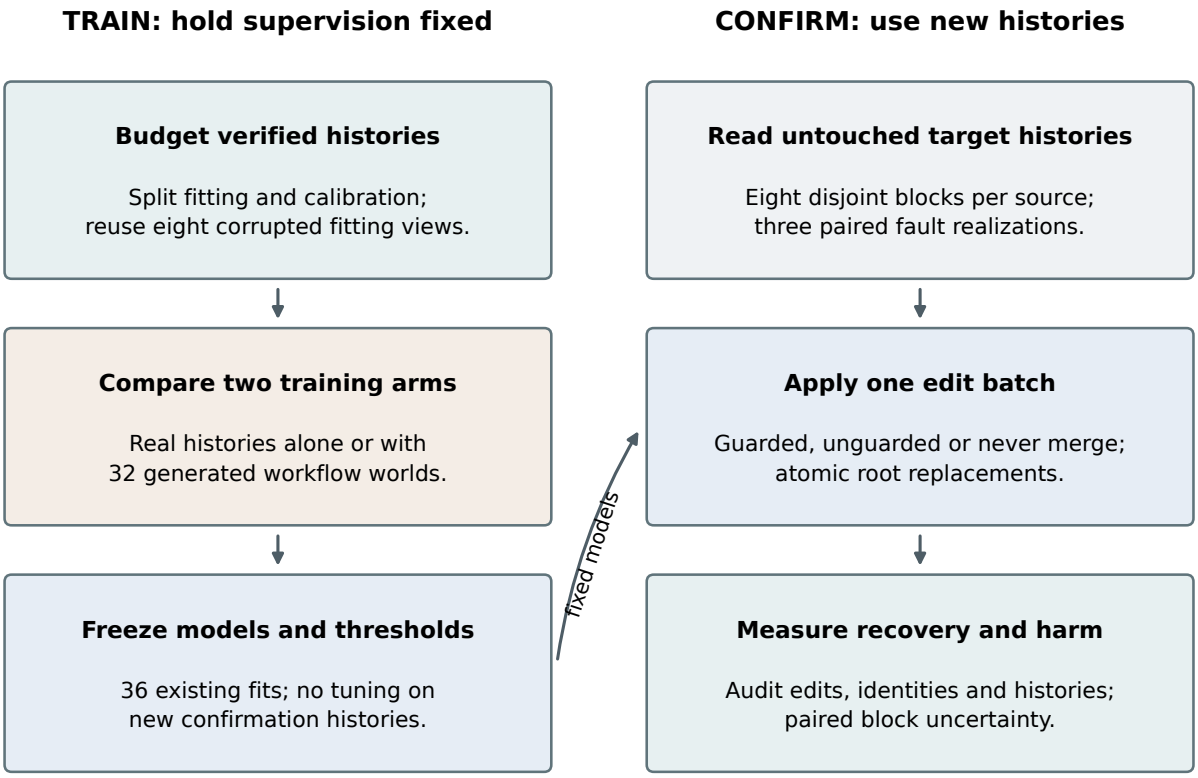


Figure 1: Confirmation design. The two arms have identical real verification budgets; generated workflows are the added supervision. Fixed models and thresholds are evaluated on untouched target histories. Additional merge-policy controls apply at 64 bundles and $q = .99$. Reference identities and fault locations remain outside inference.

The earlier decoder selects a commuting batch, applies it, recomputes observed evidence and repeats for at most four passes. The repeated-corruption study and its confirmation use one pass (Section 4.4). A batch cannot simultaneously merge an entity and edit its participation. Accepted participation edits are not reversed; accepted merges cannot be split by inference. Every batch retains a reversible audit record. For operational graphs, candidate limits are 128 participants per event/type and 64 identity candidates per anchor. Candidate coverage is reported separately from scoring accuracy.

4 Experimental design and verification budgets

4.1 Public sources and disjoint target histories

Loan data come from BPI 2017 [16]. Targets are applications and explicitly referenced offers; resources are context. Purchasing uses the released BPI 2019 OCEL representation [17, 18, 19]; targets are purchase orders and items, with resources and vendors as context. Identifiers and copied trace/object fields are removed from learner attributes. The adapters retain allowed event-level loan descriptions and purchasing cumulative net worth. An absent native offer link is not proof that no

Table 1: Real reference access for the repeated-corruption models reused in confirmation. Ranges cover three fixed allocation seeds; both training arms receive the same histories. Event and target-entity counts cover fitting plus calibration. Artificial corruptions add no new real histories. These counts measure simulated reference access, not human time.

Source	Total bundles	Fit/calibration	Events	Target entities
Loans	16	12/4	586–637	33–38
Loans	64	48/16	2,248–2,320	146–151
Loans	256	192/64	9,072–9,213	590–599
Purchasing	16	12/4	226–300	54–65
Purchasing	64	48/16	1,005–1,679	280–351
Purchasing	256	192/64	4,501–5,188	1,109–1,158

real-world relationship existed.

The original revision excluded feasibility prefixes, earlier confirmation roots and their target children. Roots sharing target children or crossing excluded target children were also excluded. A declared hash ranking assigned 512 development, 1,024 annotation and 1,024 confirmation roots per source, followed by a native-method block prospectively extended to 2,048 roots. Target children stay with their root; context resources and vendors may overlap. The current confirmation takes the next 1,024 eligible roots after all those blocks, with earlier raw selection hashes verified unchanged. Its 40,273 loan and 25,040 purchasing events have not supplied earlier fitting, calibration or repair outcomes. These are fresh target histories in known source processes.

4.2 Count fitting and calibration together

A verification unit is a complete root history bundle: the root, its target children, all their alias handles and event memberships. Total budgets of 16, 64 and 256 bundles allocate 12/48/192 to fitting and 4/16/64 to calibration. Hash assignments are nested, and a root keeps its role as the budget increases. Three allocation/fault seeds define the fixed fits. Each role's graph contains only its selected histories and their observed context. The two arms receive byte-identical real fitting and calibration arrays. Table 1 counts their reference access; historical research labels are additional costs outside these incremental adaptation budgets.

The R_8 arm fits eight damaged views of the same selected histories: mild/severe mixed errors, independent errors, missing-dominant and wrong-dominant faults, fragmentation and pipeline faults. Calibration uses three separately seeded mixed, independent and pipeline views of disjoint bundles. Each view samples at most 256 fitting or 512 calibration event queries, plus retrieved identity pairs. Query subsampling does not reduce the counted complete-history budget. Artificially corrupted views use the same underlying reference memberships; they do not represent additional investigator annotations. Missing positive or negative classes cause operation-specific abstention.

The $+G$ arm adds a fixed collection of 32 executable worlds, with 512 roots and 4,507 events. Overlapping arrivals, resource queues, lognormal service times, coarse recording clocks and varying descriptions include 262 cancellations and 728 retry events. The eight fault variants are applied after execution. Two additional empty-history graphs admit multiple equally compatible identities and teach abstention; this is a narrow constructed ambiguity, not independently observed uncertainty. Generator components are not individually ablated.

Both arms use 200 boosting iterations, 15 leaves, learning rate .08, regularization 1 and no early stopping. Augmented fitting initially gives real and generated examples equal weight mass, followed by class balancing. Thus the arms match real verification, not total examples or compute. For each

operation, calibration chooses the highest-recall threshold reaching empirical precision at least q, with at least five correct actions. Both fixed targets $q = .95, .99$ are retained; infeasible operations abstain. No confirmation outcome selects weights or thresholds. Earlier single-corruption, zero-budget and calibration-only studies use different protocols and are identified separately.

4.3 Measure useful recovery and introduced errors

For reference entity q, let H_q be its event history, A_q its observed aliases, and $\widehat{H}(u)$ the events retrieved by the predicted cluster of handle u. We measure history retrieval by

$$F_{\text{history}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|A_q|} \sum_{u \in A_q} \frac{2|\widehat{H}(u) \cap H_q|}{|\widehat{H}(u)| + |H_q|}.$$

Entities with no reference events do not enter Q. Exact-history recovery replaces the inner score by an equality indicator, retaining the same entity/alias weighting. It does not require every alias of an entity to succeed simultaneously.

We separately measure identity-pair precision/recall and changes to initially correct events. The revised damage measure expands each predicted cluster to *all* its reference identities before checking participation. Consequently, a false merge can damage events on both sides, including events whose representative handle did not change. Direct assignment damage is also retained for comparison with the earlier study. Identity fragmentation alone can reduce history F1 while canonical event participation remains correct; a true merge can therefore improve history F1 without increasing the fully recovered-event count. This is why the never-merge comparison also reports true identity pairs. Event activity, timestamp and fields are checked to remain unchanged.

Application queries ask whether a retrieved history contains recorded cancellation evidence and whether its first/last timestamps are recovered. Cancellation evidence uses the case-insensitive substring `cancel` consistently in predictions and references. It does not establish the final business outcome, and boundary timestamps do not certify every simultaneous boundary record. Earlier studies report three-seed means and ranges. The current confirmation uses paired block uncertainty as defined below; events are never treated as independent statistical replications.

4.4 Prospective confirmation and merge abstention

We tested the follow-up's conditional findings on previously unused target histories, with no additional training or threshold selection. Before selecting these histories or computing repair outcomes, a separate protocol fixed all 36 full-information models, both calibration targets, the budget grid, hypotheses and analysis. The cohort construction above excludes every earlier target-history partition. No new organization, source schema or natural-error annotation is introduced.

A fixed second hash divides each source into eight disjoint blocks of 128 complete root histories, matching the exploratory evaluation size. All records of each selected root are retained, and each block is repaired separately. Inference context therefore remains 128 roots; full-log scaling is outside this experiment. Each block receives three new mixed-fault realizations, paired with the three previously fitted model seeds. Every budget and training arm sees the identical damaged graph within a block/seed. The complete experiment has 576 guarded budget/threshold conditions, plus 192 safety conditions at budget 64 and $q = .99$. No-edit outcomes are retained for all 48 block/seed inputs. No additional generated examples or verified fitting/calibration histories are supplied.

All confirmation policies use one simultaneous edit pass with atomic root substitutions. The safety comparison reuses identical candidates, scores and calibrated thresholds. Guarded decoding rejects tied additions and permits a merge only at score at least .99, without co-occurring handles,

and when it is mutual best with a margin of at least .05 over alternatives. The *never-merge* policy prohibits all alias merges while preserving the guarded policy's tied-addition rejection and atomic root substitutions. It can still repair participation and can still introduce wrong links. An unguarded policy retains the atomic-root contract but removes the additional confidence, margin and co-occurrence checks. Thus the comparison tests whether conservative merging adds value beyond complete merge abstention.

Two directional hypotheses were fixed. H1 predicts that augmentation at 64 bundles and $q = .99$ increases loan full-event recovery without increasing mean new damage. H2 predicts that guarded merging retains true identity pairs and improves history F1 or recovery over never merge without increasing mean new damage, tested separately for each source/arm. For uncertainty, we first average the three fixed model/fault runs within each block, then bootstrap eight paired blocks with 10,000 resamples. We report conditional 95% percentile intervals. They describe new-history variation for these fixed models in these processes; they do not treat events or seeds as independent populations. Interval support requires a positive lower bound for the gain and a nonpositive upper bound for damage. Secondary comparisons have no multiplicity adjustment.

4.5 Supporting published-method comparison

The native comparison independently masks each event's single root identifier with probability .30 in each 2,048-root cohort, using three fixed seeds. Activity, resource and time are available; child identifiers, vendor relations and descriptive fields are removed for every method. Trusted remaining case IDs provide closed-case supervision. HERE's author Transformer uses 512-dimensional states, three layers, eight heads, 15-event windows, batch size eight and 30 epochs. The final epoch is evaluated. Only retained identifiers enter training targets; overlapping-window validation is diagnostic, not an independent test. The headless runner and masking/evaluation protocol are explicit adaptations. No expert rules are supplied, so the comparison is named *HERE-Transformer without expert rules*, not a reproduction of the complete published system.

Matched controls include both fixed generated scorers, a target-local boosted scorer and nearest-known-event matching. Generated scorers choose one candidate for each missing ID. The local boosted control fits at most 4,096 retained-ID queries with leave-query-out profiles, whereas HERE uses the remaining known events throughout its training windows; this unequal fitting volume is reported. All retained memberships are observed context. Existing IDs remain unchanged. Native results therefore assess missing-ID completion, not robustness to wrong existing IDs, coupled alias repair or verification-budget efficiency. Data, fixed generated weights, implementation and all six native conditions were pinned before native outcomes; newly fitted checkpoints were recorded afterwards. The native comparison finished before the separate 156-job combined-repair protocol was frozen. That declaration retained both generated mixtures and all budget comparisons, with no subsequent model or threshold selection from confirmation outcomes.

5 Results and application interpretation

5.1 Untouched-history confirmation

The prespecified loan result replicates (Table 2). At 64 verified bundles and $q = .99$, augmentation raises mean full-event recovery from 30.2 to 62.5 per block and reduces new damage from 2.08 to 0.00. After averaging the three fixed model/fault runs within each block, recovery improves in all eight blocks. Fitted-seed sensitivity remains: the three seed-specific recovery differences, averaged across blocks, are 5.5, -11.5 and 103.0 events. Thus one fitted-seed mean loses recovery; block

Table 2: Untouched-history confirmation: means over eight disjoint 128-root blocks and three fixed fitted-model/fault seeds. R_8 repeatedly corrupts the same verified histories; $+G$ adds generated training. Budgets include calibration. R counts fully recovered events and H newly damaged events per block; all false-merge effects are included. History F1 is percent.

Source	Budget	q	F1 R_8	F1 $+G$	R R_8	R $+G$	H R_8	H $+G$
Loans	16	0.95	80.0	80.0	65.8	58.5	6.38	2.21
	16	0.99	79.8	79.9	55.6	32.4	3.71	0.42
	64	0.95	82.0	80.2	226.6	98.2	2.96	1.25
	64	0.99	79.9	79.6	30.2	62.5	2.08	0.00
	256	0.95	81.1	80.7	110.4	94.1	1.38	1.71
	256	0.99	80.1	80.0	42.2	34.2	0.75	3.58
Purchasing	16	0.95	82.8	80.8	390.0	273.4	16.54	15.00
	16	0.99	80.5	80.7	271.0	255.8	7.88	12.21
	64	0.95	83.5	82.3	565.2	455.4	15.79	26.62
	64	0.99	81.0	80.3	400.3	329.2	5.42	16.62
	256	0.95	81.3	80.4	414.7	369.3	6.42	6.92
	256	0.99	79.6	77.9	309.3	207.5	1.25	1.33

uncertainty does not cover new training draws. The paired recovery difference is 32.3 events [95% interval 29.5, 35.3], and the damage difference is -2.08 [-3.75, -0.75]. H1 meets both its directional and interval criteria. However, history F1 decreases by 0.23 percentage points: its paired difference is -0.23 [-0.30, -0.18]. The confirmed benefit concerns recovery and harm at this operating point, not uniform improvement in every endpoint.

Table 2 also retains the failures. At 256 bundles, augmentation lowers mean history F1 and recovery and increases mean damage in both sources at both targets; not every damage interval excludes zero. On purchasing at 64 bundles it reduces recovery and increases damage at both targets. Its small purchasing F1 gain at 16 bundles and $q = .99$ accompanies fewer full-event recoveries and more damage. Thus generated data can add value beyond repeated corruption, but the evidence supports a narrow operating condition rather than a general verification-budget saving or a universal rule for when to stop generating data.

The never-merge control changes the safety interpretation (Tables 3 and 4). For real-only loans, guarding reduces damage from 38.12 to 2.08, while recovery changes from 30.17 to 30.21. Never merging further reduces damage to 0.29 with the same 30.2 recoveries, but gives up 2.75 true merged pairs and some history F1. On purchasing, guarded real-only/augmented policies retain 9.46/16.08 true pairs per block. Their history F1 exceeds never merge, but their damage is 5.42/16.62, versus 4.71/15.38. The corresponding extra-harm intervals are positive. None of the four H2 comparisons establishes a beneficial merge policy without added harm over never merge.

The strict augmented loan model makes no merges under any of the three policies; all three produce identical outcomes. Its zero observed harm therefore does not demonstrate an incremental guard benefit. Moreover, real-only never merge still recovers 30.2 events versus 62.5 for the augmented model, so the recovery advantage is not explained solely by false merges in the real-only comparator. Never merging eliminates false identity merges by construction, but participation repairs can still damage records. The choice between retaining fragmented histories and accepting residual merge risk remains an application decision; the measured frontier supplies evidence, not a deployment safety guarantee.

All 768 saved repair programs passed independent replay, exact undo, event-content preservation and recomputed history/identity and cluster-expanded damage checks. The source-selection audit

Table 3: Merge-policy confirmation at 64 verified bundles and $q = .99$. R and H are fully recovered and newly damaged events. TP/FP count true/false merged identity pairs. Never merge retains the guarded participation rules and atomic root replacements. Counts and F1 (percent) average the same 24 runs per setting.

Source	Arm	Policy	F1	R	H	TP	FP
Loans	R_8	Unguarded	80.0	30.2	38.12	5.21	3.17
		Guarded	79.9	30.2	2.08	2.75	0.38
		Never merge	79.6	30.2	0.29	0.00	0.00
	$+G$	Unguarded	79.6	62.5	0.00	0.00	0.00
		Guarded	79.6	62.5	0.00	0.00	0.00
		Never merge	79.6	62.5	0.00	0.00	0.00
Purchasing	R_8	Unguarded	81.0	400.0	10.54	11.17	1.75
		Guarded	81.0	400.3	5.42	9.46	0.33
		Never merge	80.5	400.9	4.71	0.00	0.00
	$+G$	Unguarded	80.5	327.2	27.54	21.42	3.17
		Guarded	80.3	329.2	16.62	16.08	0.38
		Never merge	79.4	331.0	15.38	0.00	0.00

Table 4: Prespecified contrasts at 64 bundles and $q = .99$: mean differences [conditional 95% intervals]. The first row tests augmentation with guarded decoding; other rows subtract never merge from guarded decoding. F1 differences are percentage points; recovery and harm are events per 128-root block. Intervals resample eight blocks after averaging three fixed model/fault runs; no multiplicity correction is applied.

Source	Contrast	Δ F1	ΔR	ΔH
Loans	$+G - R_8$	-0.23 [-0.30, -0.18]	32.3 [29.5, 35.3]	-2.08 [-3.75, -0.75]
Loans	R_8: guard–never	0.27 [0.20, 0.35]	0.0 [0.0, 0.0]	1.79 [0.54, 3.38]
Loans	$+G$: guard–never	0.00 [0.00, 0.00]	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]
Purchasing	R_8: guard–never	0.52 [0.42, 0.63]	-0.6 [-1.0, -0.2]	0.71 [0.25, 1.17]
Purchasing	$+G$: guard–never	0.85 [0.76, 0.93]	-1.8 [-2.8, -1.0]	1.25 [0.62, 1.87]

verified all older partition hashes and complete coverage of the declared new roots. No new fitting or threshold search was performed. Complete per-block/per-seed outcomes, conditional intervals and measured evaluation costs accompany the manuscript.

5.2 Exploratory repeated-corruption study

The precursor study fitted 60 models and evaluated 156 repair conditions, including the full-input models reused for confirmation and the input ablations below. Choices were frozen before the new model outcomes. Evaluation uses 128 hash-selected roots from each previously evaluated confirmation cohort: 4,942 loan and 2,792 purchasing events. These reused source histories make this an exploratory follow-up, not independent confirmation. A single simultaneous proposal pass commits root replacements atomically. Both fixed calibration targets, .95 and .99, are reported. A guard rejects tied additions and permits merges only at score at least .99, without co-occurring handles, and with mutual-best scores separated from alternatives by .05. The same guard applies to both training arms. Its observed harm must still be measured; score thresholds are not guarantees. Different cohort size, context and decoding make these scores non-comparable to the earlier four-pass results.

Table 5: Repeated-corruption follow-up: three-seed means on 128 reused roots per source. R_8 uses eight corruptions of each verified fitting set; $+G$ adds generated histories. q is the calibration precision target. Recovery and damage count fully corrected initially incorrect events and newly incorrect initially correct events, with alias expansion. Equal q does not guarantee equal test damage.

			History F1 (%)		Recovered		Damaged	
Source	Budget	q	R_8	$+G$	R_8	$+G$	R_8	$+G$
Loans	16	0.95	79.2	79.4	64	67	10.0	4.0
	16	0.99	78.9	79.2	55	35	6.7	1.3
	64	0.95	81.6	79.6	237	108	1.7	1.7
	64	0.99	79.2	79.0	30	67	0.0	0.0
	256	0.95	80.5	80.0	123	102	2.3	2.3
	256	0.99	79.4	79.2	46	38	1.7	1.0
Purchasing	16	0.95	82.7	80.9	388	257	25.0	19.3
	16	0.99	80.5	81.0	269	236	7.7	13.0
	64	0.95	83.5	82.6	501	423	15.7	10.0
	64	0.99	81.1	80.5	343	309	2.7	4.7
	256	0.95	81.3	80.7	350	347	7.7	6.7
	256	0.99	79.7	78.2	253	196	1.7	1.3

Table 5 gives the complete budget/threshold comparison. A useful conditional result occurs on loans at budget 64 and $q = .99$: generation raises mean fully recovered events from 30.3 to 67.0 with zero observed new damage in either arm. It improves recovery in two seeds and loses in one; mean history F1 is slightly lower. At $q = .95$, the same augmentation reduces recovery from 236.7 to 108.3 at equal mean damage. Purchasing at budget 64 trades fewer recoveries for less damage at .95, but loses on both measures at .99. At budget 256, augmentation lowers mean history F1 in both sources at both targets. This fixed mixture therefore offers no established general advantage beyond repeated corruption; neither a universal verification-budget saving nor the optimal point to stop using generation follows from these trials.

Table 6 removes information before retrieval and refitting. On loans, removing timestamps reduces mean full-event recovery from 108.3 to 7.7, also reducing damage. Purchasing loses history F1 without context objects or descriptions. Descriptions are not uniformly beneficial: removing them slightly improves loan F1 and recovery while increasing damage. Removing all history edges yields no repairs. These results identify information dependencies of the scorer and candidate pipeline; they do not establish superiority of a graph neural architecture.

At budget 64 and .99, the guard reduces real-only loan damage from 39.7 to 0.0 events while preserving 30.3 recoveries, but reduces correctly merged identity pairs from 5.3 to 1.7. For augmented purchasing, damage falls from 10.3 to 4.7 with recovery changing from 307.7 to 308.7; some false pairs remain. All 192 empty-history model–event decisions remain unresolved both with and without the guard, so this challenge establishes no incremental guard benefit. Guarded augmented purchasing still damages 5.3 events on clean inputs and recovers only 54.0 events while damaging 39.3 under pipeline faults. Pipeline faults appear in training; this is a known-family stress test.

On a Threadripper PRO 5955WX with four fitting threads per worker, budget-64 real-only/augmented fitting takes mean 3.5/5.7 seconds for loans and 4.6/7.1 for purchasing. The shared generated feature cache costs 10.1 seconds; full-input inference including candidate construction and metric computation takes about 33/21 seconds per loan/purchasing test graph, excluding file writes. These are measured host timings, not human verification savings. All 60

Table 6: Input ablations of $+G$, budget 64, $q = .95$: three-seed means. Information is removed before retrieval and fitting. No-context retains target root/child incidence; no-history removes incidence from the evidence index. These are information ablations, not architecture comparisons.

	Loans			Purchasing		
Input	F1 (%)	Recovered	Damaged	F1 (%)	Recovered	Damaged
All information	79.6	108	1.7	82.6	423	10.0
No timestamps	78.4	8	0.0	81.7	424	7.3
No context objects	79.4	81	2.0	80.9	327	8.3
No descriptions	79.9	137	4.7	80.2	358	12.0
No history edges	78.3	0	0.0	76.2	0	0.0

calibration computations and 156 saved repairs were replayed; independent checks verified history metrics, cluster-expanded harm, event preservation and exact undo. Per-seed ranges and full costs remain in the artifact; the artifact includes the fixed recovery–damage frontier.

5.3 Earlier evidence and comparator sensitivity

The earlier paired observation study held executions and faults fixed while varying visible generated fields. Varied supervision raised mean history F1 from .791 to .805 on loans and from .809 to .826 on purchasing; a separate 96-world generated confirmation improved from .759 to .800. Limits were also visible: the loan pipeline condition decreased F1 in all three trials, and real-source training outperformed generated transfer on the BPI 2012 check. The full paired table and failures remain in the artifact.

The earlier combined-repair experiment evaluated 156 declared conditions on 1,024-root cohorts, using one real corruption view and four decoding passes. On purchasing, the varied mixture with 64 verified bundles reached 85.6% history F1 versus 80.6% for real-only training, winning all three paired trials; it also exceeded the 84.5% real-only mean at 256 bundles. It raised exact-history recovery from 36.5% to 48.7% and fully recovered 3,880 initially incorrect events versus 2,992, while damaging 237 versus 411 previously correct events. These are conditional gains against that single-corruption baseline, not evidence of a fourfold reduction in human verification time. The new repeated-corruption comparisons in Tables 2 and 5 directly address the stronger comparator.

Earlier application queries also showed why history F1 alone is insufficient. Purchasing cancellation-evidence F1 increased from 89.3% to 91.1%, but on loans improved cancellation retrieval coexisted with reduced first-timestamp accuracy. One calibration-only 16-bundle loan run damaged 9,028 initially correct events. Tightening calibration from .95 to .99 reduced damage and recovery in all four tested source/regime combinations. Every earlier method introduced some damage across its clean-input trials, and generated mixtures could increase damage under pipeline faults. Full earlier budget, operational-query, threshold and stress tables remain available with their original protocols; their different context, generator and decoder prevent direct score comparisons with the current confirmation.

5.4 Compatible published-method comparison

The evaluation masks each event's case identifier independently with probability .30 and restricts predictions to the closed catalog of known cases. Candidate recall averages 98.1% (range 98.1–98.2%) for loans and 86.0% (range 85.9–86.0%) for purchasing. Generated supervision has no universal advantage: both generated methods exceed nearest-event accuracy on loans, while gener-

Table 7: Native missing-identifier accuracy across three seeds. Entries report mean (min–max) percent.

Method	Loan (BPI 2017)	Purchasing (BPI 2019)
Generated, original	89.8 (89.5–90.1)	8.3 (7.8–8.9)
Generated, varied	88.9 (88.4–89.2)	6.1 (5.5–6.7)
Known-label HGB	94.5 (94.4–94.6)	65.1 (64.7–65.7)
Nearest event	83.7 (83.3–84.0)	59.0 (58.8–59.3)
HERE-Transformer (no expert rules)	67.9 (66.5–69.0)	42.0 (41.5–42.3)

ated accuracy falls to 8.3% and 6.1% on purchasing, well below nearest event and the known-label HGB control. The known-label HGB control has the highest accuracy in both domains. HERE-Transformer is evaluated without expert rules. This separately frozen comparison restricts every method to activity, resource and time and measures only missing-case completion. Generated-only loan accuracy does not carry over to purchasing. Candidate coverage exceeds 85% in both sources, so retrieval failure alone cannot explain the purchasing losses. The strongest control learns from trusted retained case IDs. Its advantage does not establish superiority for wrong existing links or fragmented identities, which this experiment does not introduce. The supporting artifact retains earlier examples of recovered cancellation evidence and newly damaged short histories as illustrations.

6 Scope, limitations and reproducibility

The earlier paired study supports observation variation as a useful generated-data condition. Its purchasing budget advantage is relative to a single-corruption real-only control. The fresh repeated-corruption confirmation supports the prespecified strict-loan recovery/harm benefit, while rejecting general augmentation superiority and a cost-free merge-safety benefit. These studies are not general sample-complexity results, measured reductions in human work, or evidence of transfer to an unseen institution. The repeated-corruption models use a different bounded generated collection, selected-history context and one-pass decoding from the earlier study; differences between studies cannot be attributed solely to corruption count. Fixed mixture weighting and individual generator mechanisms were not separately optimized or ablated.

All operational error labels in the main evaluation come from controlled perturbations of released representations. Public natural-error candidates were audited without substituting their tasks: S2AND identity disagreements largely require splitting overmerged profiles, while the public object-centric imperfection demonstration compares object–object relationships across logs. Neither provides independent natural business-error ground truth compatible with the implemented operations. Field validation with verified operational corrections remains open.

Sparse or empty histories may not contain enough information to identify a missing participant. False merges can contaminate many retrieved histories. Confidence thresholds fitted on a few bundles are uncertain, and precision measured on those bundles is not a deployment guarantee. The fault families, uncorrupted context, available catalog and offline window all limit transfer claims. Sampling complete roots also changes concurrency relative to a complete enterprise log.

The original study included a BPI 2012 check [15], architecture controls and generated compositional tests. Their results and failures remain in the separate supporting artifact. Four originally declared external simulation sources could not be retrieved; their planned evaluations remain explicitly unexecuted. No substitute was chosen based on outcomes. The revision preserves that

earlier study and does not present it as untouched confirmation for newly selected methods.

Recipes, split rules, label-access tests, source/model hashes, predictions, edit audits and per-trial results support reproduction. The earlier 156 combined-repair outputs passed saved-edit replay, event-content preservation and independent metric checks. The native audit reconstructs truth from pinned source memberships. Both domains' original revision partitions were rebuilt byte-identically. Refitting six earlier budget-64 models reproduced calibration decisions and thresholds, with score differences below 1.4×10^{-13}; model-file byte identity did not reproduce. The exploratory and fresh-history studies retain separate frozen protocols, per-trial outcomes and resource measurements. The fresh-history study reuses existing fits: its uncertainty analysis is conditional on those models and the two known source processes. Stored-model replay is distinct from fresh retraining; no human annotation-time study or independent hardware replication is claimed.

7 Conclusion

Untouched-history confirmation supports one prespecified benefit of generated supervision beyond repeated corruption of verified histories: greater loan event recovery with less observed harm at a strict, limited-budget operating point. At 256 bundles, the fixed augmentation has worse mean history F1, recovery and damage in both sources at both targets. It also loses on the main purchasing comparisons. Conservative identity checks substantially reduce damage relative to unguarded merging, but never merge exposes a remaining tradeoff between complete histories and new errors. A strong repair evaluation therefore needs a competitive corruption baseline, a merge-abstention control, and separate recovery and harm endpoints. Natural-error validation, new institutions and measured verification effort remain necessary to establish field value.

References

[1] F. Zetzsche et al. *Case ID Revealed HERE: Hybrid Elusive Case Repair Method for Transformer-Driven Business Process Event Log Enhancement*. Business & Information Systems Engineering 67, 311–337, 2025. https://doi.org/10.1007/s12599-025-00935-5.

[2] S. Dissegna, C. Di Francescomarino and M. Ronzani. *Graph-Based Event Log Repair*. CAiSE 2026, 423–440. https://doi.org/10.1007/978-3-032-28117-3_24. Accessible author preprint: https://arxiv.org/abs/2508.05145.

[3] M. Matyash, A. Gal and A. Senderovich. *DDTR: Diffusion Denoising Trace Recovery*. 2025. https://arxiv.org/abs/2510.22553.

[4] Q. Wang, L. Zhang, R. Cao, N. Guo, H. Zhang and C. Liu. *A Process Tree-Based Incomplete Event Log Repair Approach*. Information 16(5), 390, 2025. https://doi.org/10.3390/info16050390.

[5] T. Rekatsinas, X. Chu, I. F. Ilyas and C. Ré. *HoloClean: Holistic Data Repairs with Probabilistic Inference*. 2017. https://arxiv.org/abs/1702.00820.

[6] W. L. Hamilton, R. Ying and J. Leskovec. *Inductive Representation Learning on Large Graphs*. Advances in Neural Information Processing Systems 30, 2017. https://arxiv.org/abs/1706.02216.

[7] B. Knopp, M. Pourbafrani and W. M. P. van der Aalst. *Discovering Object-Centric Process Simulation Models*. ICPM 2023. https://ieeexplore.ieee.org/document/10271944/.

[8] A. Swevels, R. Dijkman and D. Fahland. *Inferring Missing Entity Identifiers from Context Using Event Knowledge Graphs*. BPM 2023, 180–197. https://doi.org/10.1007/978-3-031-41620-0_11.

[9] S. Sadeghianasl et al. *Object-centric event-data imperfection patterns*. Information Systems 141, 102766, 2026. https://doi.org/10.1016/j.is.2026.102766.

[10] M. Comuzzi, J. Ko and F. M. Maggi. *A Language to Model and Simulate Data Quality Issues in Process Mining*. ACM J. Data Inf. Qual. 17(2), Article 6, 2025. https://doi.org/10.1145/3743144.

[11] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba and P. Abbeel. *Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World*. 2017. https://arxiv.org/abs/1703.06907.

[12] V. Kothapalli, R. Ranjan, V. Hudovernik, V. P. Dwivedi, J. Hoffart, C. Guestrin and J. Leskovec. *PluRel: Synthetic Data unlocks Scaling Laws for Relational Foundation Models*. 2026. https://arxiv.org/abs/2602.04029.

[13] Y. Wang, J. You, C. Shi and M. Zhang. *Relational In-Context Learning via Synthetic Pre-training with Structural Prior*. 2026. https://arxiv.org/abs/2603.03805.

[14] S. Gupta, S. Manchanda, S. Bedathur and S. Ranu. *TIGGER: Scalable Generative Modelling for Temporal Interaction Graphs*. AAAI 36(6), 6819–6828, 2022. https://doi.org/10.1609/aaai.v36i6.20638.

[15] B. F. van Dongen. *BPI Challenge 2012*. Dataset, 2012. https://doi.org/10.4121/uuid:3926db30-f712-4394-aebc-75976070e91f.

[16] B. F. van Dongen. *BPI Challenge 2017*. Dataset, 2017. https://doi.org/10.4121/uuid:5f3067df-f10b-45da-b98b-86ae4c7a310b.

[17] B. F. van Dongen. *BPI Challenge 2019*. Dataset, 2019. https://doi.org/10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1.

[18] S. Khayatbashi, O. Hartig and A. Jalali. *Transforming Event Knowledge Graph to Object-Centric Event Logs: A Comparative Study for Multi-dimensional Process Analysis*. ER 2023, 220–238. https://doi.org/10.1007/978-3-031-47262-6_12.

[19] S. Khayatbashi and A. Jalali. *BPI Challenge datasets—OCEL 2.0*. Zenodo, version 1, 2026. https://doi.org/10.5281/zenodo.22663565.